
PRACTICAL WEEK 2

Task 1: Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

1) Create a file

```
saichaitanya@GeethzzSai:~$ cd pweek2
saichaitanya@GeethzzSai:~/pweek2$ nano task1.c
saichaitanya@GeethzzSai:~/pweek2$ |
```

Then press enter to open command

2) write a code

```
GNU nano 7.2 main.c
#include <fcntl.h>
#include <unistd.h>
#include <stdio.h>
#include <stdlib.h>

int main() {
    int source, destination;
    char buffer[1024];
    ssize_t bytesRead, bytesWritten;

    source = open("source.txt", O_RDONLY);

    if (source < 0) {
        perror("Error opening source file");
        return 1;
    }

    destination = open("destination.txt",
                      O_WRONLY | O_CREAT | O_TRUNC, 0644);

    if (destination < 0) {
        perror("Error opening destination file");
        close(source);
        return 1;
    }

    printf("Source file opened successfully.\n");
    printf("Destination file opened successfully.\n");

    while ((bytesRead = read(source, buffer, sizeof(buffer))) > 0) {
        bytesWritten = write(destination, buffer, bytesRead);

        if (bytesWritten != bytesRead) {
            perror("Error writing to destination file");
            close(source);
            close(destination);
            return 1;
        }
    }
}
```

```

        close(destination);
        return 1;
    }
}

if (bytesRead < 0) {
    perror("Error reading source file");
} else {
    printf("File contents copied successfully.\n");
}

close(source);
close(destination);

printf("Files closed successfully.\n");

return 0;
}

```

- 1) To save press control O
- 2) To come back to shell press control X
- 3) Then DO compilation

```

saichaitanya@GeethzzSai:~$ cd pweek2
saichaitanya@GeethzzSai:~/pweek2$ nano task1.c
saichaitanya@GeethzzSai:~/pweek2$ gcc task1.c -o task1
saichaitanya@GeethzzSai:~/pweek2$ |

```

OUTPUT:

```

saichaitanya@GeethzzSai:~$ cd pweek2
saichaitanya@GeethzzSai:~/pweek2$ nano task1.c
saichaitanya@GeethzzSai:~/pweek2$ gcc task1.c -o task1
saichaitanya@GeethzzSai:~/pweek2$ echo "Hello from Task 2 sample file" > sample.txt
saichaitanya@GeethzzSai:~/pweek2$ cat source.txt
Hello, this is a system call demonstration.
This file will be copied using open, read, write and close.

saichaitanya@GeethzzSai:~/pweek2$ ./task1
File copied successfully.
saichaitanya@GeethzzSai:~/pweek2$ |

```

Task-2: Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

```
File copied successfully.
saichaitanya@GeethzzSai:~/pweek2$ echo "Hello from Task 2 sample file" > sample.txt
saichaitanya@GeethzzSai:~/pweek2$ cat sample.txt
Hello from Task 2 sample file
```

```

root@kali:~# cat sample.txt
saichaitanya@GeethzzSai:~/pweek2$ strace -e trace=operat,read,write,close cat sample.txt
openat(AT_FDCWD, "/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libselinux.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libgcc_s.so.1", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libm.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\3\0>\0\1\0\0\0\250\2\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/usr/lib/x86_64-linux-gnu/libpcr2-8.so.0", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\0\0\0\0\0\0"... , 832) = 832
close(3) = 0
openat(AT_FDCWD, "/proc/filesystems", O_RDONLY|O_CLOEXEC) = 3
read(3, "nodev\tsysfs\nnodev\ttmpfs\nnodev\tbd"... , 1024) = 442
close(3) = 0
openat(AT_FDCWD, "/proc/mounts", O_RDONLY|O_CLOEXEC) = 3
read(3, "none /usr/lib/modules/6.18.33.2-"... , 1024) = 1024
read(3, "e 0 0\ndevpts /dev/pts devpts rw,"... , 1024) = 1024
read(3, "relatime 0 0\nmqqueue /dev/mqueue"... , 1024) = 1024
read(3, "n/user/1000 tmpfs rw,nosuid,node"... , 1024) = 104
close(3, "") = 0
close(3) = 0
openat(AT_FDCWD, "/proc/self/maps", O_RDONLY|O_CLOEXEC) = 3
read(3, "5cabf0035000-5cabf0127000 r--p 0"... , 1024) = 1024
read(3, "63fe000-7b73063ff000 r--p 000a90"... , 1024) = 1024
read(3, "00 08:30 13650 "..." , 1024) = 1024
read(3, " /usr/lib/x86_64-lin..." , 1024) = 1024
read(3, "0 13644 /us..." , 1024) = 566
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/uucore/en-US.ftl", O_RDONLY|O_CLOEXEC) = 3
read(3, "# Common strings shared across a"... , 4080) = 4080
read(3, "", 32) = 0
close(3) = 0
openat(AT_FDCWD, "/usr/share/coreutils/locales/cat/en-US.ftl", O_RDONLY|O_CLOEXEC) = -1 ENOENT (No suc
h file or directory)
openat(AT_FDCWD, "sample.txt", O_RDONLY|O_CLOEXEC) = 3
Hello from Task 2 samplr file
close(5) = 0
close(4) = 0
close(3) = 0
+++ exited with 0 +++
saichaitanya@GeethzzSai:~/pweek2$

```